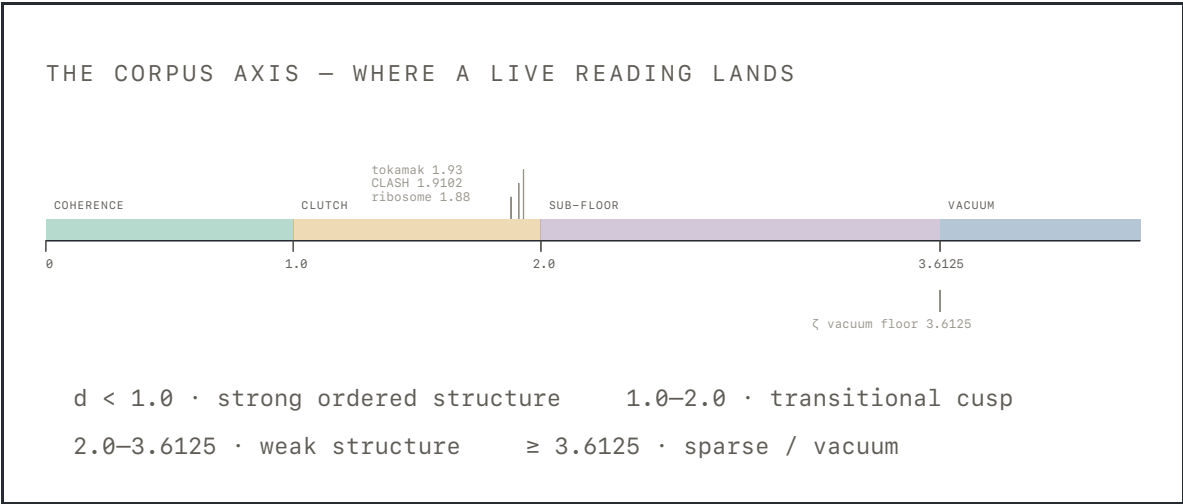


One operator, one axis, every feed.

The Resonance Layer does not score earthquakes and space weather on separate scales. It runs a single frozen operator over any ordered positive sequence and returns one number, `dij`, that lands on the same axis as every other reading in the SUPT corpus — ribosome translocation, tokamak transitions, the zeta vacuum floor. The operator has been locked since 2025-01-20 and has no tunable parameters.



01 What the number claims to be

Sheppard's Universal Proxy Theory treats measurement as resonance collapse: what an instrument reports is a collapsed harmonic proxy of a source field, not the field itself. Error is misalignment in the proxy, not randomness in the world.

$$\psi_i(\Delta t, \Delta \phi, \Delta x) = \nabla \Phi \circ R^{-1}(\mathcal{P})$$

SYMBOL	ROLE
ψ_i	Observable proxy form — what instruments report

$\Delta t, \Delta \phi, \Delta x$	Mismeasurement differentials: time, phase, space
$\nabla \Phi$	Gradient of a source potential
$\mathcal{P}$	Set of measured proxy parameters
$R^{-1}(\mathcal{P})$	Inverse resonance collapse on those proxies

Three claims carry the framework: measurement systems are proxies, not the thing itself; recalibrating the proxy changes what looks solvable; and shared addresses across domains only work if **the operator is frozen**. The Sentinel implements the frozen operator. It does not implement the theory, and `d_ij` is not a claim to be ψ_i — it is a fixed address on a shared axis.

02 The pipeline, stage by stage

IN	Ordered positive sequence $x_0 = (x_1 \dots x_n), n \geq 4$ — seismic inter-event times, flare gaps, K_p , wind speed. Order is load-bearing.
01	Robust standardization $x_i = (x_{0,i} - \text{med}) / (\text{MAD} + \varepsilon), \varepsilon = 1\text{e-}12 \rightarrow$ unitless, heavy-tail safe
02	Path accumulation $\phi_k = \sum_{i=1}^k x_i$, then $g_k = \phi_{k+1} - \phi_k$
03	Unit mean absolute step $\tilde{g}_k = g_k / (\bar{g} + \varepsilon), \bar{g} = \text{mean} g_k $
04	Phase map $u_k = \cos(2\pi \tilde{g}_k) - 1$ -periodic, phase on S^1
05	EMA resonance accumulator · frozen α $C_0 = u_0$; $C_k = \alpha u_k + (1 - \alpha) C_{k-1}$, $\alpha = 0.01$

06 Tail rule → scalar

```
T = max(50, [0.2|C|]) ; d_ij = -log(mean|C_tail| + ε)
```

07 Shuffle null RUNS IN PARALLEL

```
d(s) = probe(shuffles(x0)) ; z = (d_ij - dnull) / (σnull + ε) ;  
separated ⇔ |z| ≥ 3
```

OUT Address + band + null context

```
d_ij, band, N, z, separated, short-window flag
```

A phase-aligned path drives |C| toward larger values, so **d_ij** comes out small. A wandering path averages |C| down and **d_ij** comes out large. That is the whole reading: how close the settled accumulator sits to zero magnitude, inverted by $-\log$.

03 What counts as a signal

A number on its own means nothing. Every score ships with a shuffle null built from the same gaps in random order, fixed seed, Fisher–Yates. The question the Layer actually asks is: **is this address far from a shuffle of itself?**

OUTCOME	READING
Not separated	Null. Timing looks like a random reordering of the same gaps. Valid and informative — displayed, not hidden.
Separated, low d	Ordered structure beyond shuffle.
Separated, cusp	Landing in 1.88–1.96 alone is not evidence — heavy-tailed noise reaches it ~12% of the time. Check the tails.

Short windows ($N < 50$) let the tail span most of the accumulator. The read stays regime-valid but loses decimal precision, and the Layer flags it rather than quietly rounding.

04 The aftershock control

The obvious objection to seismic structure is that it is just aftershock clustering. The control fits a temporal ETAS model with an Omori–Utsu kernel, transforms to compensator residual times, and runs the identical frozen probe on the whitened stream.

$$\lambda(t) = \mu + \sum_{t_i < t} K e^{(\alpha_e(m_i - m_0))} (t - t_i + c)^{(-p)} \rightarrow \tau_k =$$

- **Structure survives whitening** — not explained by background plus Omori triggering.
- **Structure vanishes** — raw d_{ij} was reading clustering ETAS already describes. Still a useful fast proxy.
- **Both null** — no timing structure either way. Reported as a first-class result.

Scope: this is **temporal** whitening only. It says nothing about spatial clustering, and "structure survives" should not be read as more than it is.

05 What this is not

- Not a magnitude, CME-arrival, or Kp forecast.
- Not a language model. It is a deterministic function of the numbers in the window.
- Not tunable. α , seed, tail rule, and band edges are frozen.
- Not a claim that d_{ij} is ψ_i . It is a fixed proxy address on the corpus axis.

Timing and amplitude are separate stacks. The same operator runs on seismic gaps and solar catalog gaps; magnitude and R/S/G class are read elsewhere and never folded in.

Operator frozen 2025-01-20 · $\alpha = 0.01$ · Sheppard's Universal Proxy Theory, U.S. Copyright TXu 2-468-771 · Anchors shown for context, never fitted live.

